

IOWA STATE UNIVERSITY

LAB 5 - FATIGUE REPORT

LUCAS TAVARES VASCONCELLOS

PROFESSOR

DR. VINAY DAYAL

College of Engineering

Aerospace Engineering

Aer E 4260 Design of Aerospace Structures

AMES, OCTOBER 2024

CONTENTS

Contents	i
1 Introduction	1
2 Figures	2
3 Methodology	4
3.1 Problem 1	4
3.2 Problem 2	4
4 Conclusion	6
A Appendix A	7
A.1 Rough Work	7

INTRODUCTION

Fatigue analysis plays a critical role in ensuring the structural integrity of aerospace components subjected to repetitive loading. This report focuses on two distinct fatigue problems related to aircraft structures.

In Problem 1, the objective is to determine the number of flights until failure for a missile-holding fixture, which experiences a known stress history during each flight. The analysis is based on the material's fatigue allowable properties, using S-N curves and stress-life relationships to estimate the component's lifespan.

Problem 2 involves assessing the fatigue life and determining the Margin of Safety (M.S.) for a 7075-T6 aluminum structural element. The component is subjected to a flight stress spectrum, including a required life span of 10 years with 2 flights per day. Using Miner's Rule and applying a knock-down factor, we calculate the damage per flight and determine the TOL life cycle, ultimately computing the Margin of Safety.

FIGURES

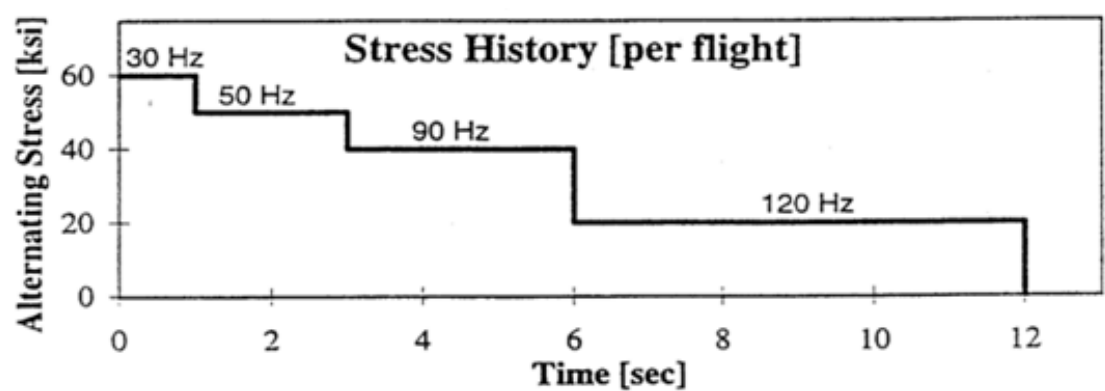


Figure 2.1: Stress history per flight for a missile-holding fixture on an aircraft (problem 1)

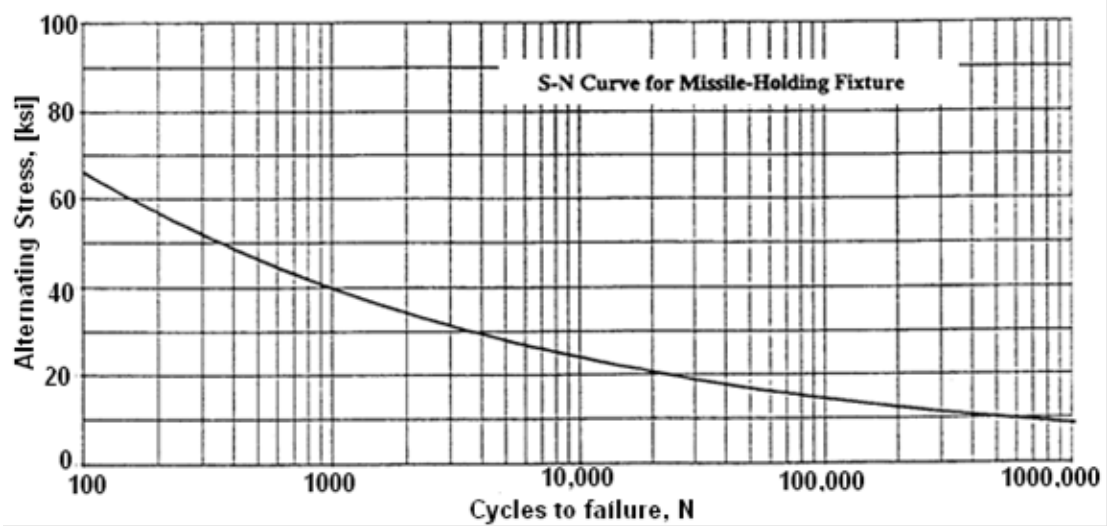


Figure 2.2: S-N curve for the missile-holding fixture (problem 1)

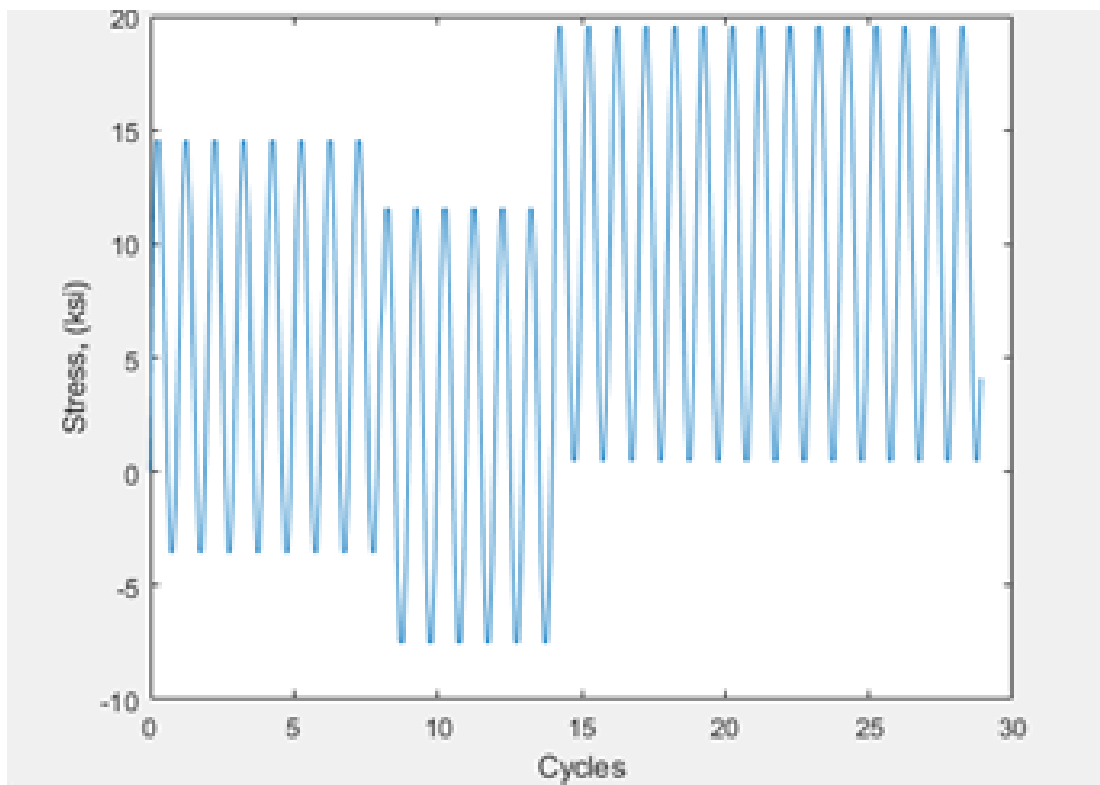


Figure 2.3: Flight stress spectrum for a 7075-T6 structural element (problem 2)

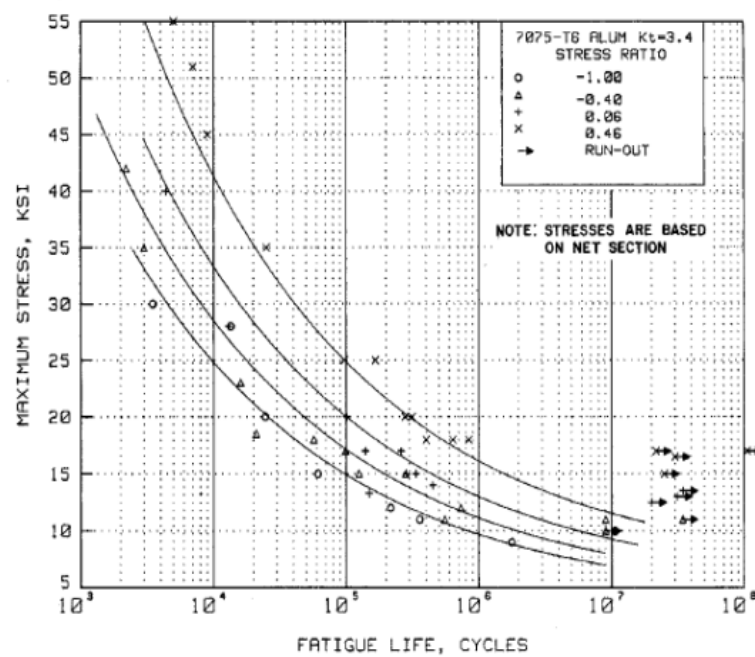


Figure 3.7.4.1.8(c). Best-fit S/N curves for notched, $K_t = 3.4$, 7075-T6 aluminum alloy rolled bar, longitudinal direction.

Figure 2.4: S-N curve for 7075-T6 aluminum alloy rolled bar (problem 2)

METHODOLOGY

3.1 Problem 1

Problem 1 was considerably straightforward to solve. From [Figure 2.1](#) we can calculate the number of cycles each alternating stress value goes through per flight (n) from the frequencies displayed. Also, from the S-N curve in [Figure 2.2](#), we can get the number of cycles for failure (N) for each alternating stress.

Alternating stress	n	N
60 ksi	30 cycles	150 cycles
50 ksi	100 cycles	350 cycles
40 ksi	270 cycles	1000 cycles
20 ksi	720 cycles	20000 cycles

With these values, we can use the Miner's rule to calculate the damage for 1 flight and, consequently, the life of the fixture.

$$\text{Damage} = \sum \left(\frac{n}{N} \right) = \frac{30}{150} + \frac{100}{350} + \frac{270}{1000} + \frac{720}{20000} = 0.792 \quad (3.1)$$

$$\text{Life} = \frac{1}{\text{Damage}} = \frac{1}{0.792} = 1.26 \Rightarrow 1 \text{ cycle} \quad (3.2)$$

The fixture would fail before the aircraft completes its second flight cycle. Thus, its life is of 1 cycle.

3.2 Problem 2

We start problem 2 by collecting the number of cycles and the maximum and minimum stress values per phase of the flight from [Figure 2.3](#). Additionally, (3.2) was used to calculate the R-ratios for each phase. With these R-ratios and the f_{max} , the number of cycles for failure for each phase was obtained from [Figure 2.4](#).

$$R = \frac{f_{min}}{f_{max}} \quad (3.3)$$

Phase	n	f_{min}	f_{max}	R	N
1	8 cycles	-4 ksi	15 ksi	-0.2667	3×10^5 cycles
2	6 cycles	-8 ksi	12 ksi	-0.6667	5×10^5 cycles
3	15 cycles	1 ksi	20 ksi	0.05	10^5 cycles

Additionally, knowing the maximum stress and R-ratio for the entire flight, we could obtain from **Figure 2.4** the Ground-Air-Ground cycle number of cycles to failure $N_{G-A-G} = 4 \times 10^4$ cycles.

With the known n and N values, we could calculate the damage.

$$\text{Damage} = \sum \left(\frac{n}{N} \right) = \frac{8}{3 \times 10^5} + \frac{6}{5 \times 10^5} + \frac{15}{10^5} + \frac{1}{4 \times 10^4} = 2.14 \times 10^{-4} \quad (3.4)$$

Applying the knock down factor of 3:

$$\text{Damage}_{\text{knock down factor}} = 3 \cdot 2.14 \times 10^{-4} = 6.41 \times 10^{-4} \quad (3.5)$$

Now, calculating the TOL cycles for failure:

$$N = \frac{1}{\text{Damage}_{\text{knock down factor}}} = 1560 \text{ cycles} \quad (3.6)$$

With that number of cycles, we can obtain the equivalent maximum stress for a TOL life cycle from **Figure 2.4**: $f_{\max} = 44$ ksi.

Knowing the life goal of 10 years with 2 flights per day, we can calculate the total number of desired cycles:

$$N = 10 \text{ years} \times 365 \text{ days/year} \times 2 \text{ cycles/day} = 7300 \text{ cycles} \quad (3.7)$$

And the maximum allowable stress for the stated life goal from **Figure 2.4** is: $F_{\max} = 27$ ksi.

Thus, we can calculate the Margin of Safety:

$$\text{M.S.} = \frac{F_{\max}}{f_{\max}} - 1 = \frac{27 \text{ ksi}}{44 \text{ ksi}} - 1 = -0.386 \quad (3.8)$$

Lastly, we can compute the actual number of flights to fatigue failure:

$$(1 + \text{M.S.}) \cdot N = (1 - 0.386) \cdot 1560 \text{ cycles} = 957 \text{ flights} \quad (3.9)$$

CONCLUSION

In conclusion, the analysis of Problem 1 indicates that the missile-holding fixture will fail after completing just one flight, as calculated using Miner's rule. In Problem 2, the structural element of 7075-T6 aluminum was found to have a negative Margin of Safety (M.S. = -0.386). This indicates that the structural element will not meet the desired life goal of 10 years with 2 flights a day, as it was calculated to have a life of 957 flights with the knock down factor of 3, which is significantly lower than the required 7,300 flights.

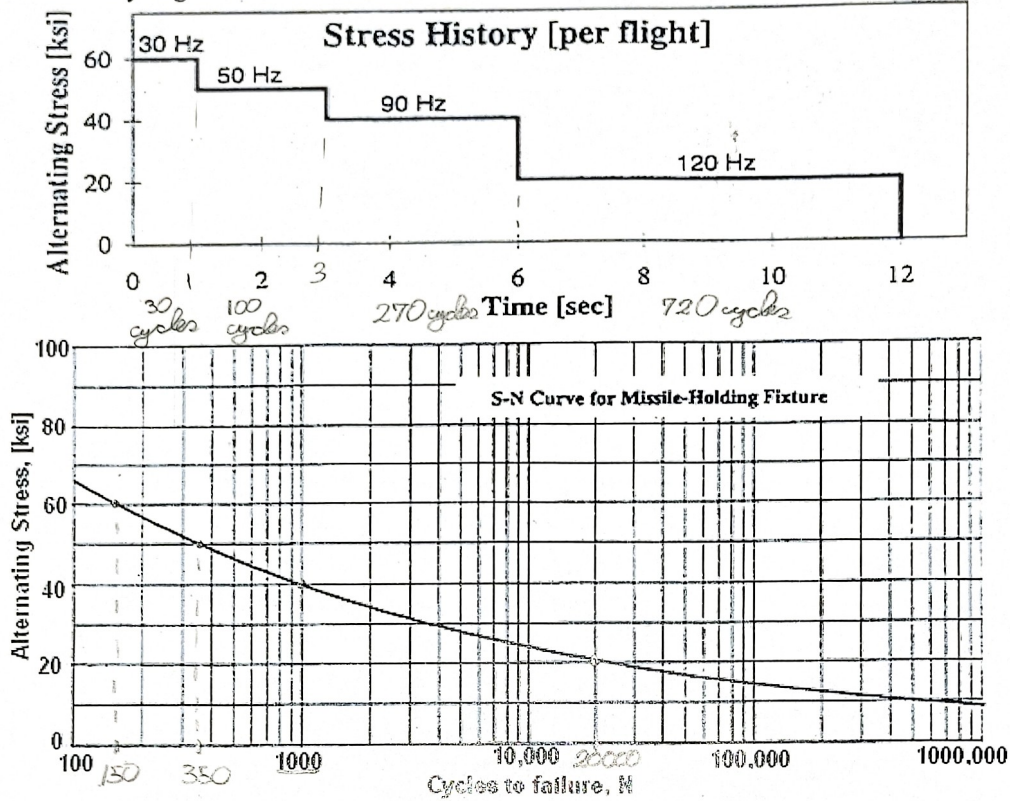
| A

APPENDIX A

A.1 Rough Work

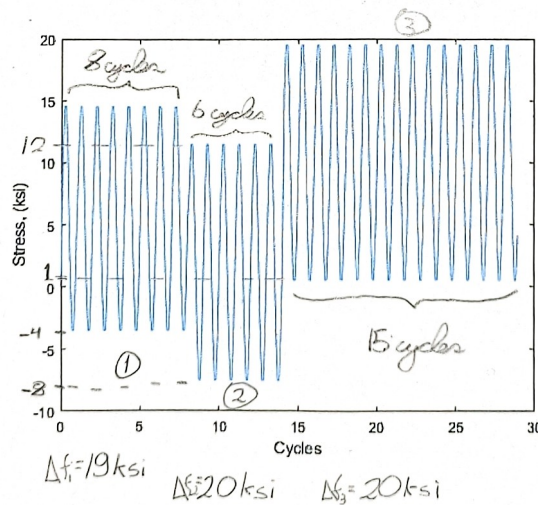
Aer E 426/AerE 526
Assignment Fatigue

Problem-1: A missile-holding fixture on an aircraft is subjected during each flight to the stress history shown below. After how many flights will the fixture fail if the material fatigue allowable is as shown.



Problem-2: The 7075-T6 structural element with $k_t=3.4$ is subjected to flight stress spectrum shown below. If the life goal is 10 years with an average of 2 flights per day, what is the Margin of Safety? The Margin of safety is given by:

$$M.S. = \frac{F}{f_{max}} - 1$$



Were F is the allowable maximum stress and f_{max} is maximum applied stress. Apply a knock down factor of 3 to reduce the "best fit" lines to "allowable lines" (95% confidence/95% reliability).

Suggested Solution Procedure:

1. Carefully count the number of cycles, n_i , for each R-ratio (f_{min}/f_{max}). Hint: Do not forget the basic G_A_G (Ground-Air-Ground) cycle.
2. Compute the damage for one flight (Takeoff-Landing, TOL) based on Miner's Rule ($\sum(n_i/N_i)$), using the allowable S-N curves to determine N_i . (Hint: The allowable S-N curves can be approximated by dividing the fatigue lives in the best fit S-N curves by the knock down factor (effectively shifting the curves to the left). Or you can multiply the damage of one TOL by 3.
3. Compute an equivalent TOL life cycle (a single cycle which causes the same damage as the actual flight spectrum). (Hint: Use the same R-ratio as the actual TOL cycle. This will result in a larger f_{max} then the actual spectrum f_{max} .)
4. Compute the maximum allowable stress for the stated life goal (using the same R-ratio as the TOL cycle and the corresponding allowable S-N curve), F
5. Compute the Margin of Safety: $M.S. = F/f_{max} - 1$
6. Compute the actual number of flights to fatigue failure (using the allowable S-N curve).

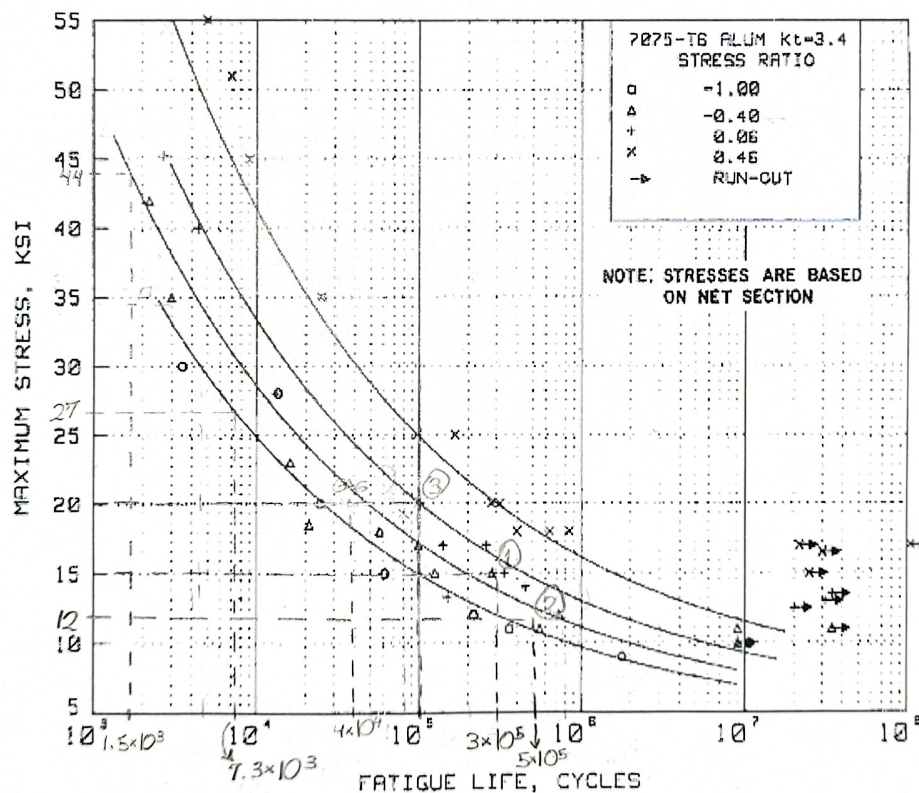


Figure 3.7.4.1.8(c). Best-fit S/N curves for notched, $K_t = 3.4$, 7075-T6 aluminum alloy rolled bar, longitudinal direction.

AERE 426 - Lab Assignment 5

$$Damage = \sum \left(\frac{m_i}{N_i} \right) \quad Life = \frac{1}{\sum \left(\frac{m_i}{N_i} \right)}$$

Problem 1). $Damage = \frac{30}{150} + \frac{100}{350} + \frac{270}{1000} + \frac{720}{2000} = 0.792$

$Life = \frac{1}{0.792} = 1.26 \text{ flights} \Rightarrow \boxed{Life = 1 \text{ flight}}$

Problem 2). $Desired \# \text{ flights} = 10y \times 365 d/y \times 24/d = 7300 \text{ flights}$

Part 1
 $m_1 = 8$
 $f_{min} = -4 \text{ ksi} \Rightarrow R_1 = \frac{4 \text{ ksi}}{15 \text{ ksi}} = 0.2667 \quad N_1 = 3 \times 10^5 \text{ cycles}$
 $f_{max} = 15 \text{ ksi}$

$m_2 = 6$
 $m_3 = 15$
 $f_{min} = -8 \text{ ksi} \Rightarrow R_2 = \frac{8 \text{ ksi}}{12 \text{ ksi}} = 0.6667 \quad N_2 = 5 \times 10^5 \text{ cycles}$
 $f_{max} = 12 \text{ ksi}$

Part 1
 $f_{min} = 1 \text{ ksi} \Rightarrow R_3 = \frac{1 \text{ ksi}}{20 \text{ ksi}} = 0.05 \quad N_3 = 10^5 \text{ cycles}$
 $f_{max} = 20 \text{ ksi}$
 $N_{GAG} = 4 \times 10^4 \text{ cycles} + 1 \text{ G.A.G. cycles}$

$\left(\sum \frac{m_i}{N_i} \right) \text{ Damage for 1 flight} = \frac{8}{3 \times 10^5} + \frac{6}{5 \times 10^5} + \frac{15}{10^5} + \frac{1}{4 \times 10^4} = 2.14 \times 10^{-4}$

Part 2 $Damage \text{ with knock down factor of } 3 = 6.41 \times 10^{-4}$

Part 6 $Actual \text{ TOL} = (1 + MS) \cdot N = (1 - 0.386) 1560 = 957 \text{ cycles}$

Part 3 $\text{with knock down factor} = 1560 \text{ cycles} \rightarrow f_{max, KDF} = 44 \text{ ksi}$

Part 4 $\text{Life goal: } 7300 \text{ cycles} \rightarrow F_{max} = 27 \text{ ksi}$

Part 5 $MS = \frac{F_{max}}{f_{max, KDF}} - 1 = \frac{27 \text{ ksi}}{44 \text{ ksi}} - 1 = -0.386$